



Performance assessment of the BDS-3 PPP-B2b service for real-time earthquake source description: a case study for the 2021 Mw 7.4 Maduo earthquake

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Abstract

Retrieving the coseismic displacement in real time is essential when the high-rate Global Navigation Satellite System (GNSS) is used for the real-time earthquake source description. In 2020, the third generation of the BeiDou Global Navigation Satellite System (BDS-3) initiated a real-time precise point positioning service through the B2b signal (PPP-B2b), which provides a new possibility for real-time earthquake source description. In this study, the performance of the PPP-B2b on the real-time retrieval of coseismic displacement was first validated with GNSS observations collected during the 2021 Mw 7.4 Maduo earthquake. Then, the warning magnitude, focal mechanism and fault slip distribution were inverted with the derived PPP-B2b displacement in a simulated real-time model. Our results indicate that the accuracies of the coseismic displacements in the north, east and up components are 0.8 cm, 0.5 cm, and 1.4 cm for the single GPS PPP-B2b, 0.5 cm, 0.6 cm, and 0.9 cm for the single BDS-3 PPP-B2b and 0.4 cm, 0.3 cm, and 0.4 cm for the combined GPS/BDS-3 PPP-B2b, respectively, when compared with the coseismic displacements of the post-processing precise point positioning (PPP). Moreover, the warning magnitude, focal mechanism and fault slip distribution inverted with the PPP-B2b displacements are all in good agreement with those inverted with the PPP displacements. This study highlights the great possibility of the PPP-B2b to be used for earthquake early warning and rapid emergency responses.

Keywords BDS-3 · PPP-B2b · Coseismic displacement · Earthquake source inversion

Introduction

A high-rate global navigation satellite system (GNSS) is recognized as an effective complementary means to traditional seismic instruments in real-time earthquake monitoring and providing early warnings, as high-rate GNSSs can directly measure the ground displacement without saturation in the

near field. Many studies have demonstrated the great performance of high-rate GNSSs on warning magnitude calculations (Crowell et al. 2013; Melgar et al. 2015; Zang et al. 2020), focal mechanism inversions (Melgar et al. 2012), fault slip distribution inversions (Allen and Alon 2011; Minson et al. 2014; Zang et al. 2022) and fault dynamic rupture process inversions (Miyazaki et al. 2004; Goldberg et al. 2020; Wen et al. 2021). It has been confirmed that both relative positioning and precise point positioning (PPP) can be used to retrieve high-accuracy coseismic displacements (Larson et al. 2003; Xu et al. 2013; Geng et al. 2017; Li et al. 2019). Compared with relative positioning, PPP can be utilized to measure ground motions from a single station alone without relying on reference stations and therefore is more suitable for earthquake monitoring. The accuracies of the PPP displacements within a short period are at the 2–4 mm and sub-centimetre levels in the horizontal and upper components, respectively, when the final precise orbit and clock products are used (Xu et al. 2013). However, when determining the final precise orbit and clock products, it has a long-time

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delay and is thus unsuitable for real-time earthquake monitoring and providing early warnings. Since 2013, the international GNSS service (IGS) has provided real-time service (RTS) and has broadcasted precise orbit and clock corrections to users through the internet in real time (Hadas and Bosy 2015), making it possible to perform real-time GNSS-based earthquake monitoring and provide early warnings.

On July 31, 2020, the third generation of the BeiDou Global Navigation Satellite System (BDS-3) began providing global service. In addition to normal positioning, navigation and timing services, BDS-3 features a satellite-based real-time precise point positioning service (Yang et al. 2021), where precise satellite orbit and clock corrections are broadcast to users through the B2b signal of BDS-3 geostationary earth orbit (GEO) satellites. Compared with RTS products, the PPP-B2b does not suffer from the influence of internet interruption, and users can obtain real-time PPP (RT-PPP) easily by decoding the received B2b corrections. The performance of the PPP-B2b in real-time positioning has been validated by a series of tests, and the results show that the positioning accuracies of the static PPP-B2b and the simulated dynamic PPP-B2b can reach centimetre and decimetre levels (Nie et al. 2021; Ren et al. 2021; Tao et al. 2021; Xu et al. 2021, 2023; Zhang et al. 2022; Tang et al. 2022; Yang et al. 2022; Sun et al. 2023). In addition to the simulated dynamic tests, Zhang et al. (2022) performed two vehicle-based kinematic PPP-B2b tests and determined that the average position errors of the combined global positioning system (GPS) and the BDS-3 are 18.6 cm and 37.1 cm in the horizontal and vertical components, respectively. Geng et al. (2022) validated the performance of the PPP-B2b in an open ocean environment and obtained results that indicate that the positioning accuracies of the combined GPS + BDS-3 in the north, east and up components are 0.047 m, 0.036 m and 0.172 m, respectively. In addition to positioning, Ge et al. (2023) investigated the method of real-time time transfer with PPP-B2b corrections. Yang et al. (2023) evaluated the performance of the PPP-B2b in real-time precipitable water vapour retrieval. In particular, Fang et al. (2022) validated the feasibility of real-time coseismic displacement retrieval with PPP-B2b by a shake table test and 2021 Mw 7.4 Maduo earthquake data. Their results indicate that the accuracies of the coseismic displacements of the PPP-B2b are 2–3 cm within a short period of time when taking the post-processed PPP displacements as a reference.

Although the performance of the PPP-B2b in real-time positioning has been validated by a series of studies, most researchers have focused on the evaluation of the PPP-B2b for a long time. In earthquake monitoring and providing early warnings, the short-term performance of the PPP-B2b matters because the energy release of an earthquake usually lasts for seconds or minutes. However, few studies, in which the performance of the PPP-B2b was assessed

for short time periods and the PPP-B2b displacement was applied for real-time earthquake source inversions, were conducted. In the study of Fang et al. (2022), only one station recorded the BDS-3 observations during the 2021 Mw 7.4 Maduo earthquake. In this study, we collected data from 13 GNSS stations that recorded both the GPS and BDS-3 observations during the 2021 Mw 7.4 Maduo earthquake. The performances of the single GPS PPP-B2b, single BDS-3 PPP-B2b and combined GPS/BDS-3 PPP-B2b for real-time coseismic displacement retrieval are all evaluated. In addition, we performed a warning magnitude calculation, focal mechanism inversion and fault slip distribution inversion with the derived PPP-B2b displacements to validate the possibility of real-time earthquake source inversion with the PPP-B2b displacements. The rest of this paper is organized as follows: First, we introduced the RT-PPP method with PPP-B2b corrections and the real-time earthquake source inversion method with PPP-B2b displacements. Then, the accuracies of the derived coseismic displacements and the inverted earthquake source parameters of the Maduo earthquake are analysed. Finally, we provided the conclusions of this study.

Methodology

In this section, the recovery procedure of the precise satellite orbit and clock offsets with the PPP-B2b corrections are first introduced. Then, the PPP-B2b positioning model and the earthquake source inversion methods are presented in detail.

Recovery of the precise satellite orbit and clock offsets with PPP-B2b corrections

At present, the PPP-B2b service mainly provides differential code bias (DCB) and satellite orbit and clock corrections to users. Although the B2b signal was designed to transmit GPS, BDS, GLONASS and Galileo corrections, only GPS and BDS-3 corrections are currently available. For GPS, PPP-B2b corrections are added to the satellite orbit and clock offsets that are calculated with the GPS LNAV broadcast ephemeris. For BDS-3, the BDS-3 CNAV1 broadcast ephemeris is adopted to calculate the initial satellite orbit and clock offsets. Then, the initial values are corrected with the PPP-B2b products. However, the satellite orbit offsets derived from the broadcast ephemeris are of the Earth-Centre-Earth-Fixed (ECEF) frame, and thus, the PPP-B2b orbit corrections, including the radial, along-track, and cross-track components, must be converted into corrections of the ECEF, which can be achieved as follows (CSNO 2020):

$$\delta X = [\mathbf{e}_{\text{radial}} \ \mathbf{e}_{\text{along}} \ \mathbf{e}_{\text{cross}}] \cdot \begin{bmatrix} \delta \mathbf{O}_{\text{radial}} \\ \delta \mathbf{O}_{\text{along}} \\ \delta \mathbf{O}_{\text{cross}} \end{bmatrix} \quad (1)$$

$$\mathbf{e}_{\text{radial}} = \frac{\mathbf{r}}{|\mathbf{r}|} \quad (2)$$

$$\mathbf{e}_{\text{cross}} = \frac{\mathbf{r} \times \dot{\mathbf{r}}}{|\mathbf{r} \times \dot{\mathbf{r}}|} \quad (3)$$

$$\mathbf{e}_{\text{along}} = \mathbf{e}_{\text{cross}} \times \mathbf{e}_{\text{radial}} \quad (4)$$

where δX represents the PPP-B2b orbit correction vector in the ECEF; $\delta \mathbf{O}_{\text{radial}}$, $\delta \mathbf{O}_{\text{along}}$ and $\delta \mathbf{O}_{\text{cross}}$ are the PPP-B2b orbit corrections in the radial, along-track and cross-track directions; and $\mathbf{r}$ and $\dot{\mathbf{r}}$ indicate the satellite position and velocity vectors calculated with the broadcast ephemeris in the ECEF. With the correction vector of δX , the precise satellite orbit offset can be recovered by:

$$\mathbf{X}_{\text{orbit}} = \mathbf{X}_{\text{broadcast}} - \delta X \quad (5)$$

where $\mathbf{X}_{\text{orbit}}$ denotes the precise satellite orbit offset and $\mathbf{X}_{\text{broadcast}}$ represents the satellite orbit offset calculated with the broadcast ephemeris.

The precise satellite clock offset can be calculated as follows:

$$t_{\text{clock}} = t_{\text{broadcast}} - \frac{C_0}{c} \quad (6)$$

where t_{clock} represents the precise satellite clock offset; $t_{\text{broadcast}}$ indicates the satellite clock offset derived from the broadcast ephemeris; C_0 is the satellite clock correction decoded from the B2b signal; and c is the speed of the light in the vacuum.

PPP-B2b positioning model

The PPP-B2b positioning model is consistent with the PPP model. Here, traditional dual-frequency ionospheric-free (IF) combinations are used to remove the influence of the first-order ionospheric delay. The code and phase observation equations of the PPP-B2b positioning model can be expressed as follows (Leick et al. 2015):

$$P_{r,IF}^s = \rho_r^s + t_r - t^s + d_{r,IF}^s - d_{IF}^s + m_r^s T + e_{r,P,IF}^s \quad (7)$$

$$L_{r,IF}^s = \rho_r^s + t_r - t^s + b_{r,IF}^s - b_{IF}^s + \lambda_{IF} N_{r,IF}^s + m_r^s T + \epsilon_{r,L,IF}^s \quad (8)$$

where $P_{r,IF}^s$ and $L_{r,IF}^s$ denote the code and phase IF combinations of the L1/L2 for GPS and B1/B3 for BDS-3, respectively; ρ_r^s represents the geometric distance from the satellite

to the receiver; t_r and t^s are the clock offsets of the receiver and satellite, respectively; $d_{r,IF}^s$ and d_{IF}^s represent the code biases of the receiver and satellite, respectively; $b_{r,IF}^s$ and b_{IF}^s are the phase biases of the receiver and satellite, respectively; m_r^s indicates the mapping function of the tropospheric delay; T is the zenith tropospheric delay; λ_{IF} and $N_{r,IF}^s$ are the wavelength and ambiguity of the IF combination, respectively; and $\epsilon_{r,P,IF}^s$ and $\epsilon_{r,L,IF}^s$ represent the noises in the code and phase observations, respectively. Other errors, such as the phase centre deviation, phase wind-up, relativistic effect, solid earth tide, ocean tide and pole tide, can be corrected with the existing models.

Notably, the PPP-B2b clock corrections refer to the L1/L2 IF combination for GPS and the B3 signal for BDS-3 (Tang et al. 2022). It is necessary to convert the BDS-3 clock correction from the B3 signal to the B1/B3 IF combination, which can be achieved as follows (Fang et al. 2022):

$$t_{B1B3}^s = t_{B3}^s - \frac{f_{B1}^2}{f_{B1}^2 - f_{B3}^2} DCB_{B1B3}^s \quad (9)$$

where t_{B1B3}^s and t_{B3}^s denote the recovered precise satellite clock offsets referring to the B1/B3 IF combination and the B3 signal, respectively; f_{B1} and f_{B3} are the frequencies of the B1 and B3 signals, respectively; and DCB_{B1B3}^s represents the DCB correction between the B1 and B3 signals decoded from the B2b signal.

As $d_{r,IF}^s$ in Eq. (7) can be absorbed by the receiver clock offset and $b_{r,IF}^s$ and b_{IF}^s in Eq. (8) can be absorbed by the phase ambiguity, the observation Eqs. (7) and (8) can be rewritten as

$$\tilde{P}_{r,IF}^s = -\mathbf{u}_r^s \cdot \mathbf{x} + \tilde{t}_r + m_{r,ZWD}^s \cdot d_{ZWD} + e_{r,P,IF}^s \quad (10)$$

$$\tilde{L}_{r,IF}^s = -\mathbf{u}_r^s \cdot \mathbf{x} + \tilde{t}_r + \lambda_{IF} \tilde{N}_{r,IF}^s + m_{r,ZWD}^s \cdot d_{ZWD} + \epsilon_{r,L,IF}^s \quad (11)$$

where $\tilde{P}_{r,IF}^s$ and $\tilde{L}_{r,IF}^s$ are the observations minus the computed code and phase, respectively; $\mathbf{u}_r^s$ denotes the unit direction vector from the satellite to the receiver; $\mathbf{x}$ is the vector of the receiver position increment; and d_{ZWD} represents the zenith wet delay. The unknown parameters, including the receiver position, clock offset, phase ambiguity and zenith wet delay, can be estimated with a Kalman filter.

Real-time earthquake source inversion with PPP-B2b displacements

Studies have confirmed that the earthquake magnitude inverted with the GNSS peak ground displacement (PGD) does not undergo magnitude saturation. The empirical

relationship between the PGD and the earthquake magnitude can be expressed as follows (Crowell et al. 2013; Zang et al. 2020):

$$\log_{10}(\text{PGD}) = A + B \cdot M_W + C \cdot M_W \cdot \log_{10}(R) \quad (12)$$

$$\text{PGD} = \max \sqrt{N_t^2 + E_t^2 + U_t^2} \quad (13)$$

where M_W denotes the earthquake magnitude; R represents the epicentre distance; A , B and C are the regression coefficients with values of -4.434 , 1.047 and -0.138 , respectively, provided by Melgar et al. (2015); and N_t , E_t and U_t indicate the real-time series of the PPP-B2b displacements in the north, east and up components, respectively.

For the focal mechanism inversion, we applied the fastCMT method (Melgar et al. 2012), where the centroid moment tensor (CMT) is inverted with the real-time static coseismic offsets derived from the GNSS displacements. In this study, we extracted the real-time static coseismic offsets by averaging the real-time PPP-B2b displacements with a 30-s sliding window. The real-time static coseismic offset d and the CMT m are related through (Melgar et al. 2012)

$$d_i^k(t) = G_{ij}^k m_j(t); (i = x, y, z; j = 1, 2, 3, 4, 5; k = 1, 2, 3 \dots, n) \quad (14)$$

where $d_i^k(t)$ refers to the i th component of the static coseismic offsets at Station k ; t represents the time epoch; m_j denotes the j th component of the CMT; G_{ij}^k is the Green's function that relates the j th component of the moment tensor to the i th component of the static coseismic offsets, which can be determined by the EDGRN program (Wang et al. 2003); and n is the number of GNSS stations.

After the inversion of the CMT at each epoch, the fault slip distribution inversion will follow shortly with the extracted real-time static coseismic offsets on the fault plane derived based on the inverted CMT results. The fault length and width are set to three times and one time the output values of the scaling laws developed by Wells and Copper-smith (1994). The entire fault plane is discretized into 8-km along-strike by 6-km down-dip independent small patches, and the slips of all patches are linearly inverted with the non-negative least-squares method. The inverse problem can be written as follows:

$$d(t) = G(t) \cdot u(t) \quad (15)$$

where d denotes the extracted real-time static coseismic offsets at epoch t ; u represents slips on each patch; and G is the matrix of the Green's function determined by the Okada mode (1985) that relates the unit slip in each patch to the permanent ground coseismic offset. To solve the ill-posed problem in the inversion, Laplacian spatial regularization is

used, and the optimal regularization parameter is determined by the L-curve method (Zang et al. 2022).

Experiment and results analysis

The 2021 Mw 7.4 Maduo earthquake occurred in the eastern portion of the Tibetan Plateau at 18:04 UTC on May 21 and was in the centre of the Baya Har block, one of the most active blocks in China. This earthquake caused obvious surface rupture and the west part of the surface rupture zone occurred along the pre-existing Kunlun Pass-Jiangcuo fault (Wang et al. 2021; Pan et al. 2021; Chen et al. 2022; Ren et al. 2022). In this study, we utilized the data from 13 GNSS stations (Fig. 1) with epicenter distance less than 200 km to evaluate the performance of the PPP-B2b in real-time earthquake source inversion. These stations can record both GPS and BDS-3 observations and are of the Crustal Movement Observation Network of China (CMONOC), the Continuously Operating Reference Stations of Qinghai Province in China (CORS) and the China Mobile Communications Group Co., Ltd. The PPP-B2b corrections are collected by an FRII-Plus receiver manufactured by Femtomes Tech (Beijing) Inc. We first processed the collected GNSS observations with the PPP-B2b positioning model in a simulated real-time model. For comparison, the GNSS observations are also processed with the traditional PPP method, where the final precise orbit and clock products provided by the International GNSS Service (IGS) Data Centre of Wuhan University are used. Then, we inverted the warning magnitude, focal mechanism, and fault slip distribution with the derived PPP-B2b displacements. To analyse the inverted results, the abovementioned earthquake source parameters are also inverted with the PPP displacements.

Accuracy assessment of the PPP-B2b displacements over a relatively long time

To evaluate the performance of the PPP-B2b, we processed the collected GNSS observations with the single BDS-3 PPP-B2b positioning model (PPP-B2bC) and the combined GPS/BDS-3 PPP-B2b positioning model (PPP-B2bGC). For comparison, we also processed the GPS/BDS-3 observations with the traditional PPP method. Four-hour displacements from 14:00 to 18:00 before the earthquake at Station QHAG are presented in Fig. 2. The displacements of the PPP are the most stable during this period. Compared with the displacements of the PPP-B2bC, the displacements of the PPP-B2bGC agree well with those of the PPP. Table 1 shows the statistical standard deviation (STD) values of the displacements of all stations. It can be found that the STD values of the PPP-B2bGC for the north, east and up components are 2.0 cm, 2.8 cm and 4.9 cm, respectively, which are larger

Fig. 1 Distribution of the collected GNSS stations (blue triangle). The red star represents the epicentre of the 2021 Mw 7.4 Maduo earthquake. The black line is the Kunlun Pass-Jiangcuo fault. The black rectangle in the upper-left inset map indicates the location of the Tibetan Plateau, where the white lines represent secondary block boundaries in and around the Tibetan Plateau and the blue rectangle is the location of the study area

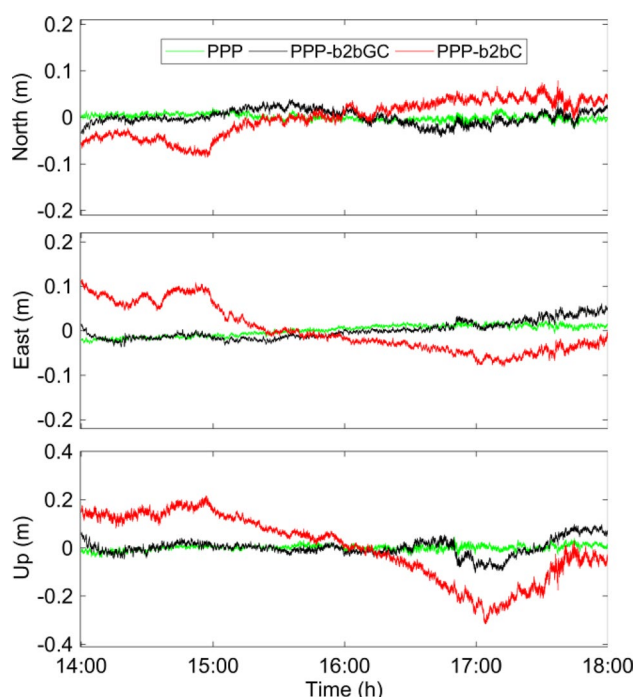
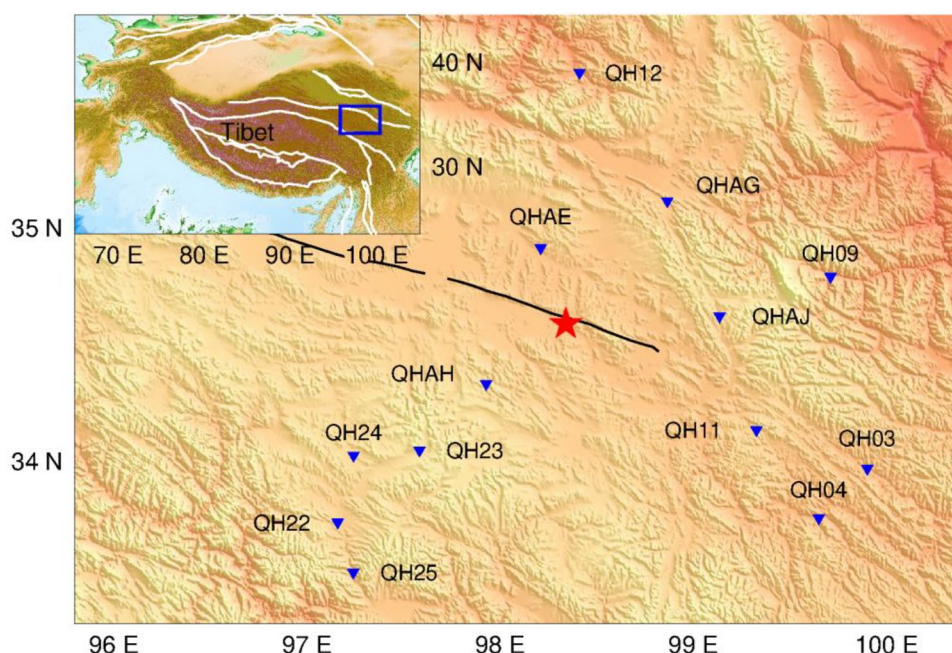


Fig. 2 Four-hour displacements derived from the PPP, combined GPS/BDS-3 PPP-B2b (PPP-B2bGC) and single BDS-3 PPP-B2b (PPP-B2bC) at Station QHAG before the 2021 Mw 7.4 Maduo earthquake

than those of the PPP with the values of the 0.7 cm, 1.3 cm and 1.5 cm. For the PPP-B2bC, the STD values in the three components are 3.6 cm, 4.5 cm and 10.9 cm. It is clear that the fusion of the GPS/BDS-3 PPP-B2b can reduce the STD values when compared with the single BDS-3 PPP-B2b. It

Table 1 Statistical standard deviation (STD) values of the four-hour displacements for the different solutions at all GNSS stations (unit: cm)

Solutions	North	East	Up
PPP	0.7	1.3	1.5
PPP-B2bGC	2.0	2.8	4.9
PPP-B2bC	3.6	4.5	10.9

should be noted that we do not present the statistical results of the single GPS PPP-B2b here, as the GPS PPP-B2b displacement causes significant errors due to the shortage of the PPP-B2b corrections for some GPS satellites.

Accuracy assessment of the PPP-B2b displacements during a relatively short time

In contrast to the long time period allowed for crustal deformation monitoring with GNSSs (Meng et al. 2019), earthquake ruptures usually last for seconds or minutes; thus, the short-term performance of the PPP-B2b is more important when the PPP-B2b is used for real-time earthquake source description. Here, we divided the four-hour displacements into short periods with a length of 5 min to evaluate the short-term performance of the PPP-B2b. The statistical STD values of all short-period displacements for the different solutions are shown in Table 2. Compared with the STD values of the four-hour displacements in Table 1, the STD values of the short-period displacements of all solutions show an obvious reduction, particularly for the PPP-B2bGC and the PPP-B2bC in the upper component. When comparing

Table 2 Statistical standard deviation (STD) values of the short-period (5 min) displacements for the different solutions at all GNSS stations (unit: cm)

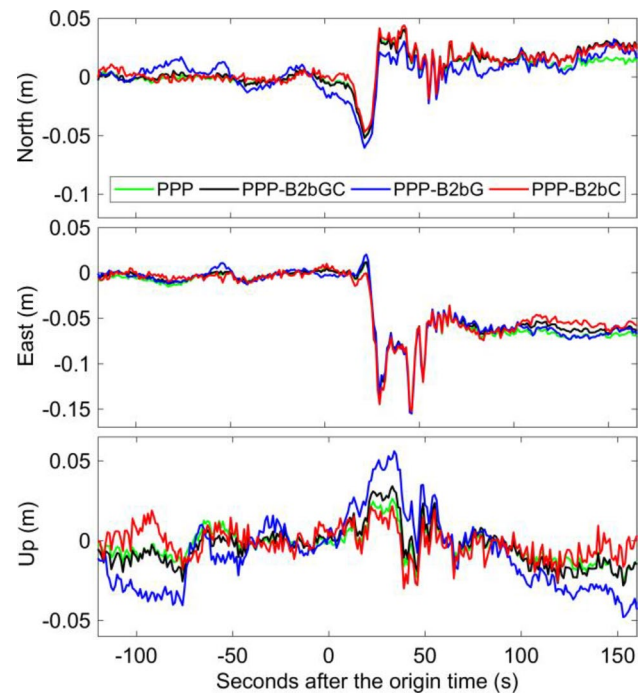
Solutions	North	East	Up
PPP	0.4	0.3	0.8
PPP-B2bGC	0.5	0.4	1.1
PPP-B2bC	0.6	0.6	1.4

the STD values between the different solutions, we found that the STD values of the PPP-B2bGC are much closer to those of the PPP in Table 2, and the STD differences between them are 0.1 cm, 0.1 cm and 0.3 cm for the north, east and up components, respectively. These results are significantly better than the STD differences between them as shown in Table 1, where the corresponding difference values are 1.3 cm, 1.5 cm and 3.4 cm, respectively. When comparing the PPP-B2bGC with the PPP-B2bC, it is clear that the STD differences between them as shown in Table 2 are superior to those in Table 1, and the difference values are 0.1 cm, 0.2 cm, and 0.3 cm and 1.6 cm, 1.7 cm, and 6.0 cm for Tables 1 and 2, respectively. All these statistical results indicate that the PPP-B2b has better performance during the short term for both the single BDS-3 and the combined GPS/BDS-3. Compared with the single BDS-3 PPP-B2bC, the combined GPS/BDS-3 PPP-B2bGC reduces the displacement noise at frequencies less than approximately 0.01 Hz and greater than approximately 0.03 Hz (Fig. S1).

Coseismic displacements derived from the PPP-B2b during the 2021 Mw 7.4 Maduo earthquake

Figure 3 shows the coseismic displacements derived from the different PPP-B2b solutions at Station QHAG. Here, we also presented the coseismic displacements derived from the single GPS PPP-B2b (PPP-B2bG). In the PPP-B2bG processing, the GPS observations were first processed with the static mode before 18:02. Then, dynamic processing was carried out to prevent the reconvergence of the displacement caused by the shortage of PPP-B2b corrections. As shown in Fig. 3, there are good agreements in the coseismic displacements between the PPP-B2bGC and the PPP for all three components. Compared with the PPP-B2bC, the PPP-B2bG shows more fluctuations, particularly in the upper component, which might be related to the GPS satellite geometry. From Fig. S3, we can find that the position dilution of the precision (PDOP) of the PPP-B2bG is larger than that of the PPP-B2bC and the PPP-B2bGC.

Taking the coseismic displacements from the PPP as a reference, we calculated the coseismic displacement errors of the other three PPP-B2b solutions by subtracting the coseismic displacements of the PPP from those of the other three PPP-B2b solutions. The derived error series of the different solutions are presented in Fig. 4. Moreover, the PPP-B2bGC has the smallest errors in all three components

**Fig. 3** Coseismic displacements derived from the PPP, PPP-B2bGC, PPP-B2bG and the PPP-B2bC at Station QHAG during the 2021 Mw 7.4 Maduo earthquake

when compared with the other two PPP-B2b solutions. The errors of the PPP-B2bG and the PPP-B2bC are comparable in the east component while the PPP-B2bG shows the relatively larger errors in the north and up components. When comparing the errors between the different stations, it seems that almost all stations show similar variations for the PPP-B2bG and the PPP-B2bC, which might be caused by the common mode error related to the satellite geometry. Table 3 lists the statistical root mean square errors (RMSEs) of the different PPP-B2b solutions, where we can find that the PPP-B2bGC achieves the highest accuracies, with RMSE values of 0.4 cm, 0.3 cm and 0.4 cm for the north, east and up components, respectively. These values are superior to the RMSE values of the PPP-B2bG and the PPP-B2bC. The RMSE values of the PPP-B2bGC are 50%, 40%, and 71% and 20%, 50%, and 56% smaller in the north, east and up components than those of the PPP-B2bG and PPP-B2bC, respectively.

Magnitude inversion with the PPP-B2b PGD

For the earthquake early warning and the rapid emergency responses, the warning magnitude can be determined with the GNSS PGD in real time, which is defined as the maximum value of the coseismic displacements from the origin time of the earthquake to the current epoch. At each station, we extracted the PGD at every epoch after the seismic wave

Fig. 4 Coseismic displacement errors of the PPP-B2bG, PPP-B2bC and the PPP-B2bGC. For clarity, the errors of the PPP-B2bC and the PPP-B2bGC are offset in the vertical axis, and the different stations are distinguished by the different colours

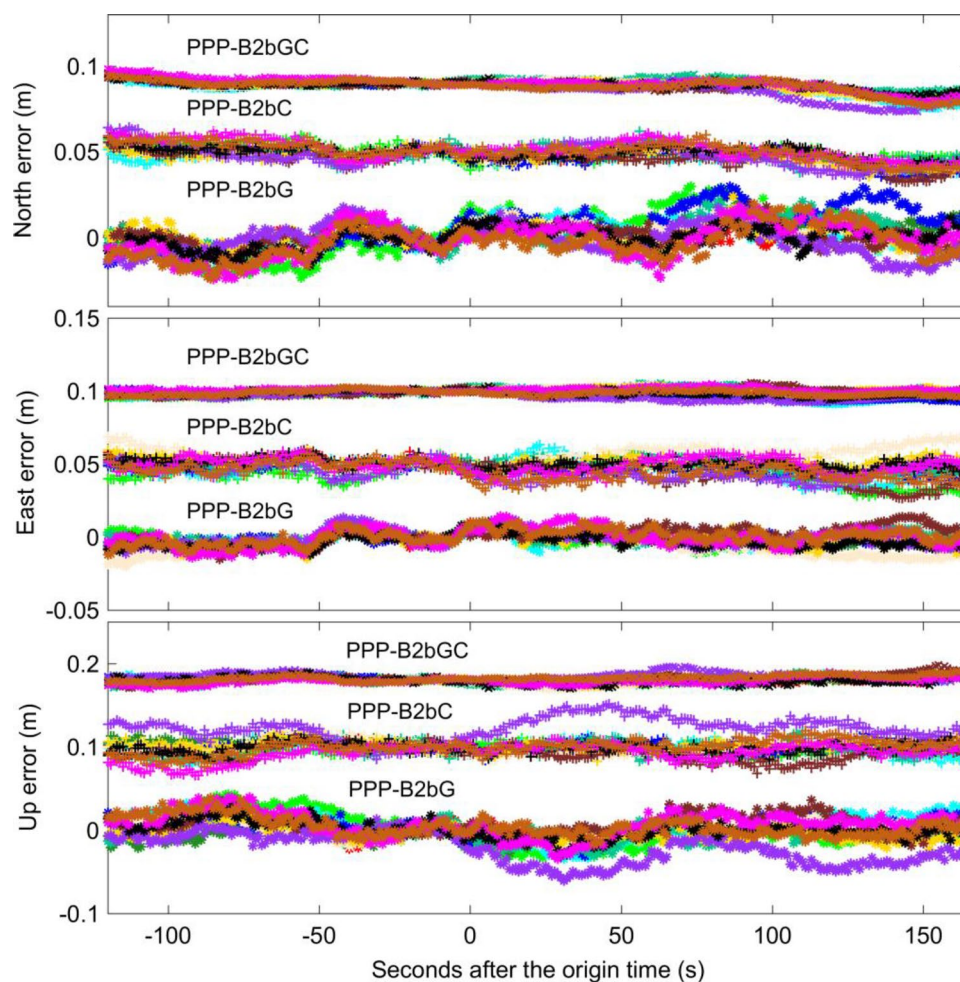


Table 3 Statistical coseismic displacement RMSE values of the PPP-B2bG, PPP-B2bC and PPP-B2bGC (unit: cm)

Solutions	North	East	Up
PPP-B2bG	0.8	0.5	1.4
PPP-B2bC	0.5	0.6	0.9
PPP-B2bGC	0.4	0.3	0.4

arrives at the station with an imaginary velocity of 6 km/s. This velocity was used because it approximately corresponds to the velocity of the P wave. The derived PGD series at each station are presented in Fig. 5, where we can find that the PGDs of the PPP, PPP-B2bG, PPP-B2bC and PPP-B2bGC are comparable at almost all times. Taking the PPP PGDs as a reference, the maximum PGD differences for the PPP-B2bGC, PPP-B2bG and PPP-B2bC are 0.7 cm, 4.5 cm and 3.9 cm, respectively. It is clear that the PPP-B2bGC has better performance than the PPP-B2bG and the PPP-B2bC.

With the derived PGDs, we calculated the warning magnitude at each epoch, which is the average value of the magnitudes determined with the PGDs at each station with the empirical relationship of (12). Figure 6 presents the time series of the derived magnitudes of the different

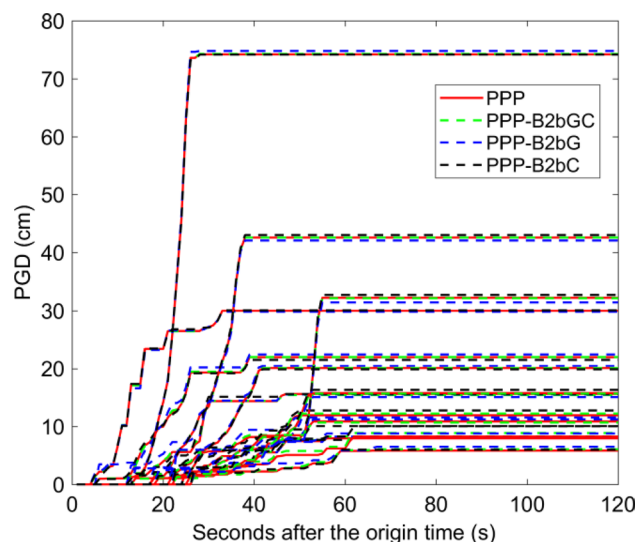


Fig. 5 PGDs extracted in real time from the coseismic displacements of the PPP, PPP-B2bG, PPP-B2bC and PPP-B2bGC for all GNSS stations

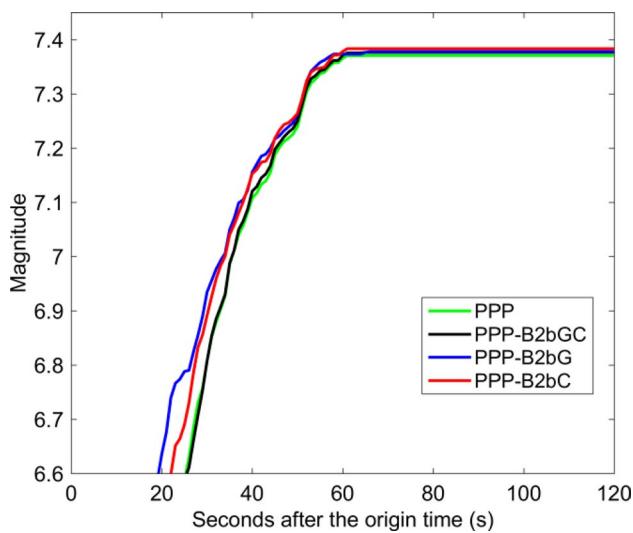


Fig. 6 Time series of the warning magnitudes determined with different GNSS PGDs

GNSS solutions. Moreover, we determined that there is good agreement between the magnitudes of the PPP and the PPP-B2bGC during the entire duration of the earthquake. For the PPP-B2bG and the PPP-B2bC, the derived magnitudes are slightly larger than that of the PPP before approximately 40 s, and these relatively larger differences decrease over time. All magnitudes tend to be stable after approximately 60 s, and the stable magnitudes of the PPP, PPP-B2bG, PPP-B2bC and PPP-B2bGC are Mw 7.37, Mw 7.38, Mw 7.38

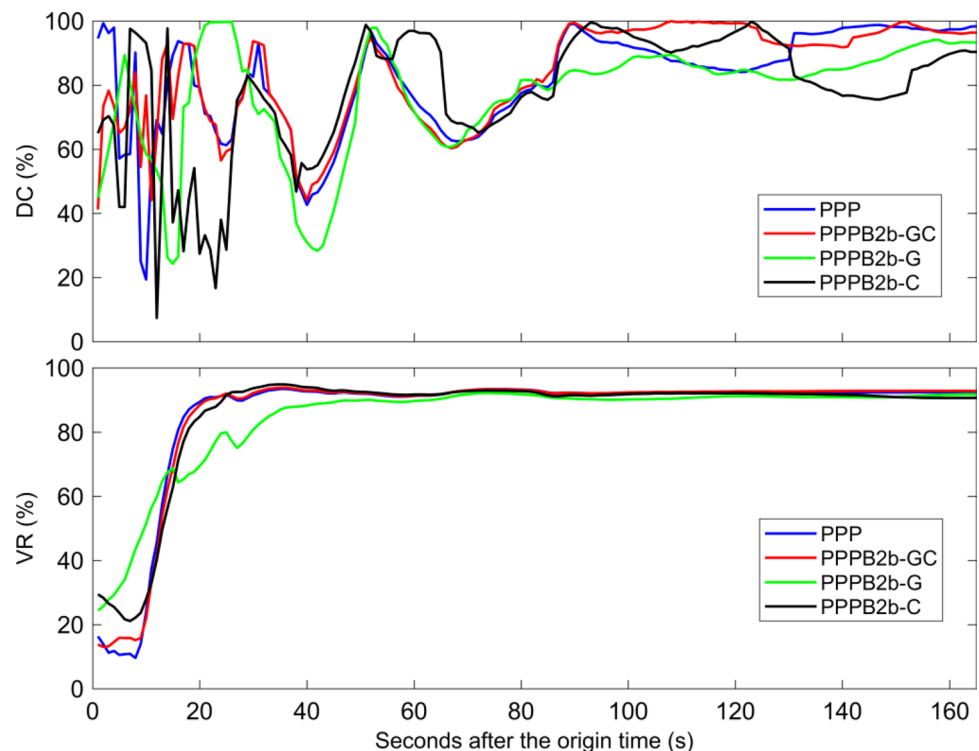
and Mw 7.38, respectively, which are all close to the final magnitude of Mw 7.4.

Centroid moment tensor inversion with the real-time coseismic offsets derived from the coseismic displacements of the PPP-B2b

Real-time and robust estimation of the centroid moment tensor (CMT) is important for assessing the characteristics of an earthquake. Here, we inverted the CMT in a simulated real-time mode with the fastCMT method. The real-time static coseismic offsets are retrieved by averaging the coseismic displacements with a 30-s sliding window, and the derived real-time static coseismic offsets of the different GNSS solutions are presented in Fig. S4-7. Considering the relatively high accuracy of the GNSS displacement in the horizontal component, the relative weight between the horizontal coseismic offsets and the up coseismic offsets is set to 3:1. In addition, the stations with epicentre distances less than 50 km are extra weighted with a value of two, given the better constraint of the near-field stations on the centroid depth (Zang et al. 2022). To determine the characteristics of the earthquake source, the inverted CMTs are further decomposed into a better double couple (DC) component and a compensated linear vector dipole component.

The percentage of the DC component and the variance reduction (VR) that illustrates the misfit between the observation and the model prediction are presented in Fig. 7, where we can find that the percentages of the

Fig. 7 Percentages of the better double couple (DC) component decomposed from the different CMT results and the variance reductions (VR) of the different CMT solutions



DC components of all solutions show similar variations after approximately 40 s, and all solutions tend to be stable after approximately 80 s. The average percentages of the DC components after 80 s for the PPP, PPP-B2bGC, PPP-B2bG, and PPP-B2bC are 92%, 95%, 86% and 88%, respectively. These high DC percentages are indicative of a good solution for a tectonic earthquake. For the variance reduction, as shown in Fig. 7, the PPP, PPP-B2bGC and PPP-B2bC show good consistency during the whole earthquake. For the PPP-B2bG, there are relatively larger differences in the VR between it and the other solutions before approximately 40 s, and these differences fade away gradually after approximately 60 s. The average VRs after 80 s for the PPP, PPP-B2bGC, PPP-B2bG, and PPP-B2bC are 92%, 93%, 91% and 92%, respectively. Both sets of results show good fits to the data.

Figure 8 illustrates the focal mechanisms derived from the different CMT results. We found that the focal mechanisms derived from the different CMT results are similar after approximately 40 s, and it can be basically determined that this earthquake was a strike-slip event after approximately 40 s based on the derived results. Table 4 compares the nodal planes derived from the focal mechanisms of the different agencies, where the results of the PPP-B2bGC, PPP-B2bG and the PPP-B2bC are the average values of the nodal planes determined after the 80 s. It is seen that the two nodal planes of the PPP-B2bGC, PPP-B2bG and the PPP-B2bC are essentially in good agreements with that of the PPP, and they are all close to the result of Zhang and Xu (2021) except the rake in Nodal Plane 1 and the dip in Nodal Plane 2.

Table 4 Nodal planes derived from the focal mechanisms of the global centroid moment tensor project (GCMT), U.S. geological survey (USGS) and this study

Solutions	Nodal Plane 1 Strike°/ Dip°/Rake°	Nodal Plane 2 Strike°/Dip°/ Rake°
GCMT	282/83/−9	13/81/−173
USGS	99/79/−38	197/53/−166
Zhang and Xu (2021)	281/88/1	191/89/178
PPP	278/80/28	183/63/169
PPP-B2bGC	279/80/34	182/56/168
PPP-B2bG	279/82/24	185/66/171
PPP-B2bC	277/83/33	183/58/172

Fault slip distribution inversion with the real-time static coseismic offsets derived from the coseismic displacements of the PPP-B2b

The fault slip distribution inversions are carried out at each epoch when the focal mechanisms are available. We inverted the fault slips on both nodal planes, and the plane with the largest variance reduction was selected as the fault plane of this epoch. In the inversion, the relative weight between the horizontal coseismic offsets and the up coseismic offsets was also set to 3:1. The fault slip distributions inverted at every epoch with the static coseismic offsets of the PPP, PPP-B2bGC, PPP-B2bG and PPP-B2bC are presented in Move. S1, Move. S2, Move. S3 and Move. S4, respectively. Furthermore, we determined that the fault planes determined with the different GNSS solutions change slightly

Fig. 8 Focal mechanisms derived from the CMT results inverted with the different static coseismic offsets. For clarity, the focal mechanisms are displayed with an interval of 10 s

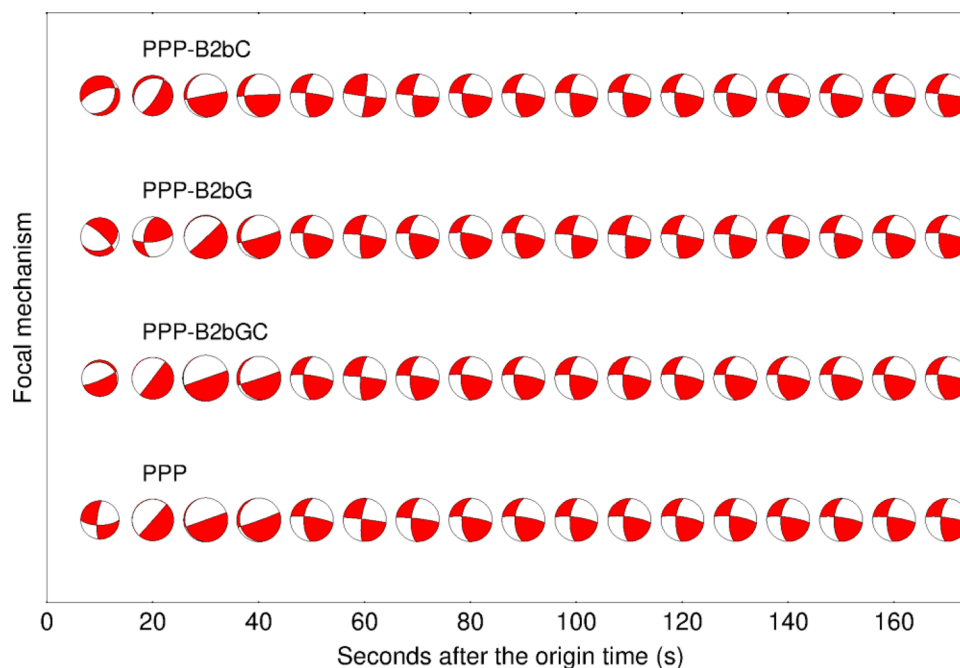
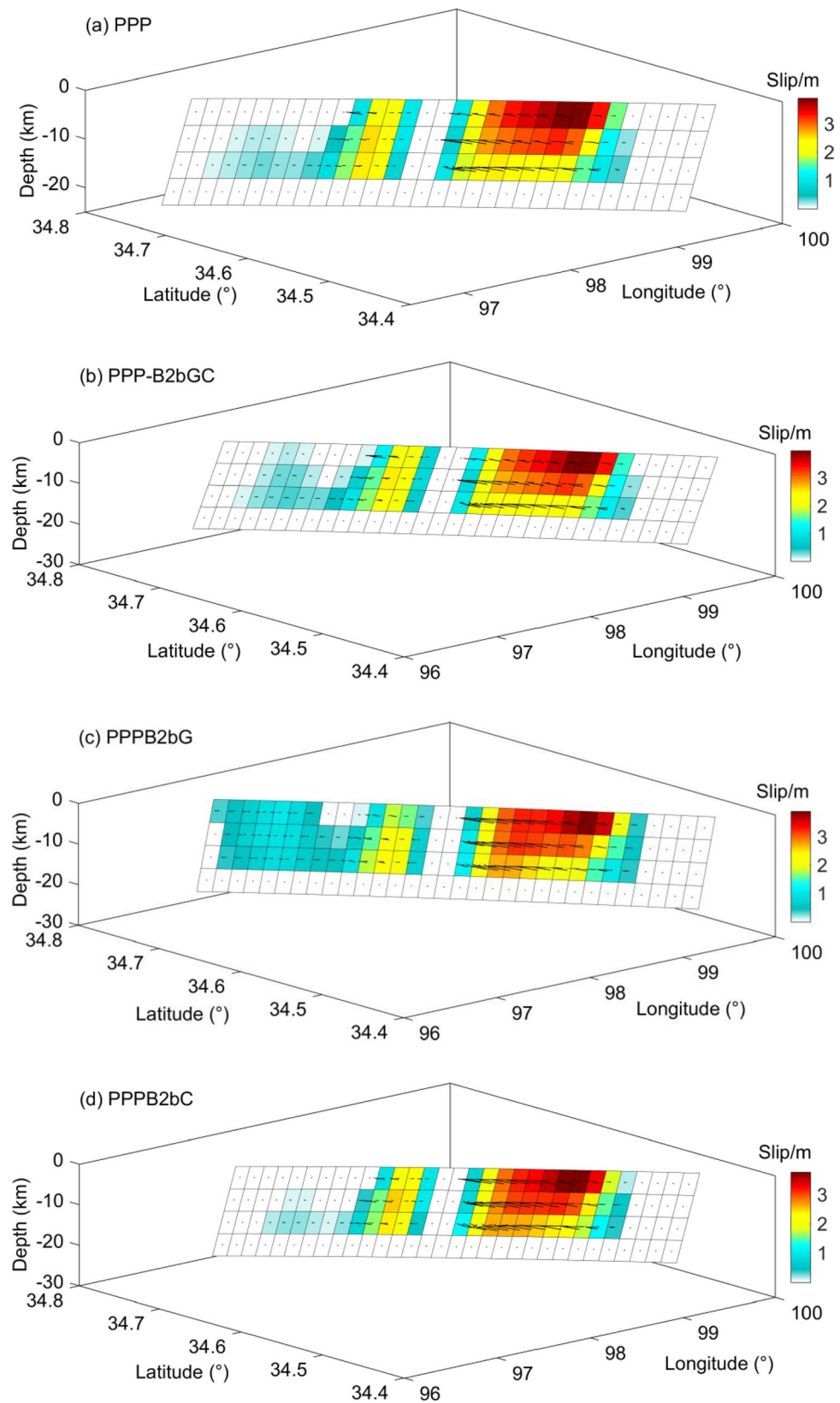


Fig. 9 Fault slip distributions inverted with the static coseismic offsets of the PPP, PPP-B2bGC, PPP-B2bG, and PPP-B2bC at 100 s after the earthquake



after approximately 40 s, and the fault slip distributions inverted with the different static coseismic offsets are also comparable. Taking the fault slips inverted with the different static coseismic offsets at 100 s as an example, we found that the fault slip distributions of the PPP-B2bGC, PPP-B2bG and PPP-B2bC are all in good agreement with that of the PPP (Fig. 9). All inverted results show that this earthquake ruptures the ground surface. There are mainly two rupture areas in the fault, and most ruptures clustered in the east segment of the fault with maximum slips of 3.9 m, 4.0 m, 3.9 m and 3.8 m for the PPP, PPP-B2bGC, PPP-B2bG and PPP-B2bC, respectively. The variance reductions of the four fault slip models are 90%, 91%, 89% and 89%, respectively, with both sets of results showing good fits to the data. The final magnitudes derived from the fault slip models of the PPP, PPP-B2bGC and PPP-B2bC are Mw 7.42, Mw 7.43 and Mw 7.42, respectively. For the PPP-B2bG, the derived magnitude is Mw 7.46, which is slightly higher than that of the other three models. This is because the static coseismic offset accuracy of the PPP-B2bG is slightly worse than those of the other solutions and leads to more small slips (Fig. 9c) in the western boundary of the fault plane that may not exist.

Conclusions

In this study, we evaluated the performance of the PPP-B2b for real-time earthquake source description with GNSS observations measured during the 2021 Mw 7.4 Maduo earthquake. The results indicate that the PPP-B2b performs better during the short term. The STD values of the PPP-B2bGC displacements during the 5 min are 0.5 cm, 0.4 cm and 1.1 cm for the north, east and up components, respectively, which are slightly larger than the STD values of the PPP displacements. Taking the coseismic displacements of the PPP as a reference, the PPP-B2bGC can be used to retrieve real-time coseismic displacements with accuracies of 0.4 cm, 0.3 cm and 0.4 cm for the three components, respectively, which are superior to those of the PPP-B2bC and PPP-B2bG. For the PPP-B2bG, although the accuracy of the horizontal coseismic displacement is less than 1.0 cm, it should be noted that the GPS observations were processed with the static mode before the earthquake. In the realistic dynamic environment, the PPP-B2bG might show poor performance due to the influence of the shortage of the PPP-B2b corrections, which might be improved with the improvement of the generation strategy of the PPP-B2b corrections in the future. For the earthquake source description, we determined that the inverted real-time warning magnitudes, focal mechanisms and fault slip distributions of the PPP-B2bGC, PPP-B2bC and PPP-B2bG are all very close to those of PPP. Thus, we conclude that the PPP-B2b has great potential to be used for providing earthquake early warning

and real-time earthquake source descriptions. However, we need to note that we only validated the possibility of the PPP-B2b being used for large earthquake source descriptions in this study. Whether the PPP-B2b has good performance on real-time earthquake source descriptions for middle or even small earthquakes needs to be validated further. In addition, the PPP-B2b corrections for GPS are incomplete and are currently unavailable for Galileo and GLONASS. With increasing PPP-B2b corrections, the accuracy and reliability of the inverted real-time earthquake source might be further improved.

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Author contributions JZ performed the data processing and wrote the manuscript. SF contributed to the experimental design, analysis and proof-read the manuscript. CX provided the advice and guidance for the results analysis. ZL contributed to the GNSS data collection and processing. RF and YL contributed to the recovery of the precise satellite orbit and clock offsets with the PPP-B2b corrections.

Data availability The final precise orbit and clock products from the Wuhan University can be obtained at <ftp://igs.gnsswhu.cn/pub/>. The GNSS observations from the CMONOC, the CORS and the China Mobile Communications Group Co., Ltd. can be obtained under permission.

Declarations

Conflict of interest The authors declare no conflicts of interest.

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